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MILITARY SPECIFICATION

RELAY, SENSITIVE

(For Torpedo Mark 37 Mods 0, 2, and 3)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification describes the requirements for the procurement, inspection, and test of a double-pole, double-throw, sensitive electromagnetic gate relay of single coil construction, Drawing 815710 or 1155715.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-S-642	Sodium Phosphate, Tribasic Anhydrous, Dodecahydrate, and Monohydrate, Technical
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-S-365	Silver Plating (Electrodeposited)
QQ-Z-325	Zinc Plating (Electrodeposited); Requirements for
TT-C-490	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

FSC 1355

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TT-I-558	Ink, Marking, Stencil, Opaque, for Nonporous Surfaces (Metals, Glass, etc.)
TT-L-58	Lacquer, Spraying, Clear and Pigmented Cellulose Nitrate (General Interior Use)
TT-P-664	Primer, Coating, Synthetic, Rust-Inhibiting Lacquer Resisting
TT-T-291	Thinner, Paint, Volatile Mineral Spirits (Petroleum Spirits)
PPP-B-621	Boxes; Wood, Nailed and Lock-Corner

Military

MIL-P-116	Preservation, Methods of
MIL-M-3171	Magnesium Alloy, Processes for Corrosion Protection of
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-P-8585	Primer; Coating, Zinc Chromate, Low Moisture Sensitivity
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-E-15090	Enamel, Equipment, Light Gray (Formula No. 11)
MIL-P-15328	Primer, Pretreatment (Formula No. 117 for Metals)
MIL-V-16399	Varnish, Moistureproof
MIL-T-18404	Torpedoes, Environmental Requirements; General Specifications for
MIL-E-82590	Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

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STANDARDS

Military

MIL-STD-108	Definitions of, and Basic Requirements for, Enclosures of Electric and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-type Base

DRAWINGS

Naval Ordnance Systems Command

Drawing 815710	Relay (Specification Control Drawing)
Drawing 1155715	Relay, Gate

PUBLICATIONS

Naval Ordnance Systems Command

OS 6341	Miscellaneous General Ordnance Design Requirements

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards Publications

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

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3. REQUIREMENTS

3.1 Description. The sensitive relay is a sensitive, electromagnetic relay of single coil construction, with a double-pole, double-throw armature. With the coil in its normally deenergized state, the armature closes the circuits through one pair of poles. When the coil is energized and the coil current reaches a predetermined value, the armature moves to close the circuits through the opposite pair of poles, at the same time opening the circuits through the initial pair of poles. When the current in an energized coil is reduced below a predetermined value, the armature will return to the position assumed when the coil is deenergized. The coil is energized from a nominal 14 volts dc or rectified ac power source.

3.2 General requirements

3.2.1 Data required. No data is required by this specification (other than reports accompanying samples submitted for preproduction and periodic production testing), or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.2.2 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the relay shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing relays that comply with the technical requirements of the contract. See 4.3.1.

3.2.3 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the relay shall conform to the detailed requirements indicated on the drawings and specifications listed in this specification.

3.2.4 Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimension for the relay shall be in strict accordance with applicable drawings, specifications, and publications listed in this specification, and shall conform to the following requirements:

3.2.4.1 General. All machine operations shall be completed before the application of the corrosion-resistant treatment.

3.2.4.2 Plating. Electrodeposited metallic coatings shall conform to the following with the plating thickness in all cases adequate to ensure conformance with the parts or equivalent specification for conductivity and corrosion resistance.

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(a) Zinc - Specification QQ-Z-325, Type II, Class 2 or 3

(b) Chromium - Specification QQ-C-320, Type I

(c) Nickel - Specification QQ-N-290

(d) Silver - Specification QQ-S-365

NOTE

Cadium plating shall not be used.

3.2.4.3 Tin and lead-tin alloys. Platings of tin and lead-tin alloys, when used, shall be of adequate thickness and so continuous as to pass the marine environment test.

3.2.4.4 Corrosion-resistant steel

3.2.4.4.1 Cleaning. Remove all oil, grease, dirt, and other foreign matter from corrosion resistant steel parts by vapor degreasing, washing with petroleum spirits (TT-T-291, Grade 1), or by immersing in a boiling solution of trisodium phosphate (O-S-642), followed by a cold water rinse and a final rinse in hot water. If necessary because of scale, immerse in the following solution, held at 140 degrees Fahrenheit (F) (60 degrees centigrade (C)) until the scale is removed:

Nitric acid, 20 percent by volume

Hydrofluoric acid, 1 to 3 percent by volume

Water, 77 to 79 percent by volume

Then rinse in cold water and blow dry with compressed air. If structures are fabricated with overlapping joints, the preferred method of cleaning is by vapor degreasing to avoid entrapment of caustic or acid solutions.

3.2.4.4.2 Passivation. Immerse the clean metal in a solution of equal parts of nitric acid (40 degrees Baume) and water, to be held at a tempature of 160 degrees F (71 degrees C) for five minutes. Rinse first in clean cold water, followed by a rinse in hot water. Then blow dry with compressed air.

3.2.4.5 Anodic film. Parts fabricated from aluminum or aluminum alloys shall be anodized in accordance with MIL-A-8625. Surfaces which are subsequently painted may be given a protective treatment in accordance with MIL-C-5541.

3.2.4.6 Magnesium and magnesium alloys. Magnesium and magnesium alloys, when authorized, shall be finished in accordance with MIL-M-3171.

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3.2.4.7 Painting. If painting is selected as the final finish, the following processes shall be used:

3.2.4.7.1 Cleaning. Surfaces to be painted shall be free of dirt, dust, grease, mill scale, welding flux, and other foreign matter. Cleaning may be accomplished with volatile grease solvent, vapor degreasing detergents, hot alkali cleaning solutions, or other processes which have no deleterious effect on the surface. Welding irregularities shall be removed. Alkali cleaning shall not be used on aluminum or aluminum alloy surfaces. When alkali cleaning solution is used, the final rinse shall consist of a diluted chromic acid solution (0.5 pounds of chromic acid per 100 gallons of water). When fabrication involves overlapping joints, vapor degreasing is the preferred method of cleaning.

3.2.4.7.2 Pretreatment. For steel surfaces to be painted, either of the following processes may be used:

(a) One coat of wash pretreatment in accordance with MIL-P-15328, applied as a continuous film 0.0004 to 0.0005 inch thick.

(b) A hot-dip phosphate or chromate-phosphate treatment conforming to TT-C-490, Grade 1.

3.2.4.7.3 Primer. One coat of primer in accordance with TT-P-664 or MIL-P-8585 (for aluminum or magnesium surfaces) applied as a continuous film 0.0004 to 0.001 inch thick.

3.2.4.7.4 Enamel. Two coats of enamel, light gray, MIL-E-15090, Class 2, applied as a continuous film approximately 0.001 inch thick.

3.2.4.7.5 Exceptions. The following surfaces shall not be painted: mounting surfaces, pins, and threaded parts.

3.2.5 Moisture, fungus, and corrosion. The relay shall be rendered moisture, fungus, and corrosion resistant as required by OS 6341.

3.2.6 Marking. Marking shall be by means of engraving, metal stamping, embossing, or permanently affixed nameplate, or by ink stencil, TT-I-558. Ink shall be black, or of a color contrasting to the surface it covers. Ink, TT-I-558, when applied to an enamel surface, shall be covered with moisture proof varnish, MIL-V-16399; when applied to other surfaces it shall be covered with lacquer, TT-L-58, Type 1.

3.2.6.1 Marking data. Unless otherwise specified in the contract or order, or detailed in the applicable drawings, marking shall include the following: NAVORD drawing number and revision letter; operating voltage; contact rating; coil resistance; manufacturer's name; symbol or trade mark; manufacturer's part or assembly number if desired; and schematic diagram.

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3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.3 Definitions. Definitions applicable to the environmental requirements of this specification are given in MIL-E-82590.

3.4 Construction and design

3.4.1 General objectives. The basic objectives are that the equipment meets the needs of the Naval service and that it reflects the maximum simplicity, reliability, ease of maintenance, and economy of production. It should also be capable of withstanding long periods of storage.

3.4.2 Dimensions and tolerances. Dimensions and tolerances of the sensitive relay shall conform to Drawing 815710 or 1155715.

3.4.3 Weight. The weight of the relay shall not exceed three ounces.

3.4.4 Mechanical design

3.4.4.1 Type. The relay shall be of the self-aligning type, and shall remain in adjustment in any position.

3.4.4.2 Adjustments. Any adjustable features of the relay shall be so designed that the setting of any adjustment will not be altered by tests or environmental conditions described herein. There shall be no external adjustments.

3.4.4.3 Enclosure. The relay shall be hermetically sealed as defined in MIL-STD-108.

3.4.4.4 Mounting. The relay shall be mounted by two number 4-40 studs located as shown on Drawing 815710 or 1155715.

3.4.5 Electrical design

3.4.5.1 Contacts. The contacts of the relay shall be of silver.

3.4.5.2 Contact rating. The rated current of the contacts shall be two amperes at 28 volts with a resistive load.

3.4.5.3 Contact arrangement. The contact arrangement of the relay shall be double-pole, double-throw, as shown in Drawing 1155715 or 815710. The schematic diagrams of the relay shown on the above drawings shall be included in the marking of the relay.

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3.4.5.4 Case. The case of the relay shall not be used for grounding nor shall it be electrically connected to any terminal of the relay.

3.4.5.5 Terminals. Eight terminals of the "J" hook type, capable of accommodating two number 18 wires each, shall be provided and located as shown in Drawing 1155715 or 815710. Terminals shall be hot tin dipped or solder tinned.

3.5 Performance requirements and product characteristics. The sensitive relay shall meet the following performance requirements and product characteristics:

3.5.1 Dielectric strength. The sensitive relay shall withstand an ac voltage of 700 volts root-mean-square (rms) at 60 Hertz (Hz) for a period of one minute between any relay terminal and the relay case, and between any coil terminal and any contact terminal without damage, arcing, or breakdown.

3.5.2 Insulation resistance. The insulation resistance between any terminal of the relay and the case, and between any coil and contact terminal, shall be not less than 100 megohms.

3.5.3 Contact resistance. The resistance across the normally closed contacts when the relay coil is deenergized, and across the normally open contacts when the relay is energized and the contacts are carrying current as in 3.4.5.2, shall not exceed 0.05 ohm.

3.5.4 Contact bounce. The gate relay shall not exhibit any characteristics of contact bounce or chatter while in either the energized or deenergized state. The duration of contact bounce shall not exceed two milliseconds during the closing of the normally open contacts when the coil is energized and during the closing of the normally closed contacts when the coil is deenergized.

3.5.5 Coil resistance. The coil resistance of the relay shall be 5000 ohms plus or minus 10 percent, when measured at a temperature of 68 degrees F plus or minus 10 percent (20 degrees C plus or minus 10 percent).

3.5.6 Coil current

3.5.6.1 Operate. The maximum current at which the normally open contacts transfer to the closed condition shall be not greater than 2.8 milliamperes (ma). Under the environmental vibration tests, the normally closed contacts shall remain closed with a coil current up to but not exceeding 2.25 ma.

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3.5.6.2 Release. The minimum current at which the normally open contacts transfer from a closed to an open condition shall be not less than one ma. Under environmental vibration tests, the normally open contacts shall remain closed with a coil current decreasing to, but not below, 1.4 ma.

3.5.7 Sensitivity

3.5.7.1 Operate. The operating sensitivity shall be a nominal 40 milliwatts.

3.5.7.2 Release. The release sensitivity shall be a nominal 5 milliwatts.

3.5.8 Operating and releasing time

3.5.8.1 Operating. The operating time shall be not greater than that shown in the accompanying curves for the condition shown thereon (Figure 1). Operate time is the duration of time between the initial closing of the coil circuit and the initial closing of the normally open contacts (contact bounce not included).

3.5.8.2 Releasing. The release time shall be not greater than that shown in the accompanying curves for the conditions shown thereon (Figure 2). Release time is the duration of time between the initial opening of the coil circuit and the initial opening of the normally open contacts.

3.5.8.3 Transcurrent switch time. The transcurrent switch time (relay swinger travel time from the opening of the normally closed contacts to the closing of the normally open contacts), when the voltage and current change rate is slower than 40 milliwatts per second, will be less than 6 milliseconds.

3.5.9 Operating life. The relay shall be capable of meeting the performance requirements and product characteristics specified herein after 100,000 contact transfers under the conditions described in 4.4.1.

3.6 Environmental requirements. The relay shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.5.

3.6.1 Vibration. The relay shall meet the following vibration requirements:

3.6.1.1 Vibration, nonoperating. The relay shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.1.2.

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3.6.1.2 Vibration, operating. The relay shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.5.1.3.

3.6.1.3 Vibration, transportation. The relay shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.1.4.

3.6.2 Shock, transportation. The relay shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.2.

3.6.3 Temperature and humidity

3.6.3.1 High temperature and humidity, nonoperating. The relay shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.3.1.

3.6.3.2 High temperature, operating. The relay shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.3.2.

3.6.3.3 Low temperature, nonoperating. The relay shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.3.3.

3.6.3.4 Low temperature, operating. The relay shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.3.4.

3.6.4 Pressure. The relay shall be subjected to the pressure requirements for conditions pertinent to surface and submarine launched weapons as specified in 4.5.4 and thereafter show no damage and meet the performance requirements and product characteristics of this specification.

3.7 Cleanup. Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

3.8 Workmanship. The relay shall be constructed and finished in a manner to ensure compliance with all requirements of this specication. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, riveting, machine screw assemblage, welding, brazing, freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

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#1 CURRENT OVERDRIVE = $\frac{\text{STEADY STATE CURRENT}}{\text{RELAY PULL-ON CURRENT}}$

#2 VOLTAGE OVERDRIVE = $\frac{\text{EFFECTIVE SOURCE VOLTAGE}}{\text{RELAY PULL-ON VOLTAGE}}$

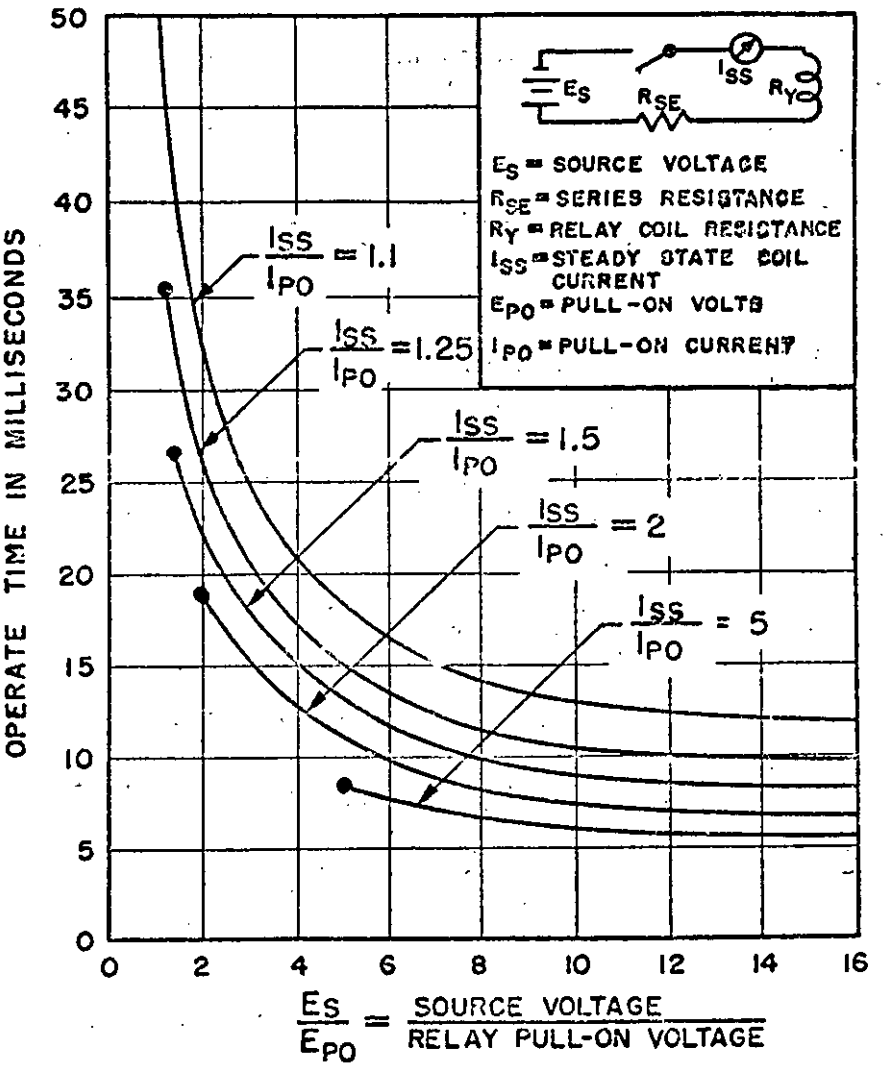


FIGURE-1

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the relays shall be classified as follows:

(a) Preproduction inspection (paragraph 4.5)

(b) Quality conformance inspection (paragraph 4.4)

(c) Periodic production inspection (paragraph 4.5)

4.3 Sampling

4.3.1 Preproduction. A preproduction sample of five relays, manufactured in accordance with 3.2.2, shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the relays by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.4, a random sample of five relays will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The relays shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

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4.4 Quality conformance inspection. Each relay delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.4.1 Test conditions. Unless otherwise specified, the relay shall be subjected to quality conformance tests under the following conditions:

4.4.1.1 Temperature: Room ambient 65 degrees F to 95 degrees F (18.33 degrees C to 35 degrees C).

4.4.1.2 Altitude: Normal ground.

4.4.1.3 Vibration: None.

4.4.1.4 Humidity: Room ambient to 95 percent relative maximum.

4.4.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, when definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required tests.

4.4.3 Test equipment. The following items of test equipment are required to perform the quality conformance tests set forth in this specification:

4.4.3.1 A source of 700 volts rms, 60 Hertz, ac power.

4.4.3.2 A 500 volt dc megohmmeter, with a midscale reading of not less than 100 megohms.

4.4.3.3 A Kelvin bridge, or equivalent.

4.4.3.4 A nominal 35-volt source of dc power.

4.4.3.5 A resistance, rated at least $1\frac{1}{2}$ watts, variable from 5000 to 50,000 ohms.

4.4.3.6 A dc voltmeter, scale 0 to 50 volts, accurate to within plus or minus 1 percent.

4.4.3.7 Oscilloscope, Tektronix Model 545A or equivalent.

4.4.3.8 Saw-tooth generator capable of delivering 30 volts peak into a 10,000 ohm load with sweep rate adjustable from 6 Hertz (Hz) to 10 Hertz.

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4.4.3.9 A dc milliammeter, scale 0 to 10 ma, accurate to within plus or minus 1 percent.

4.4.3.10 Three dc ammeters, scale 0 to 5 amperes, accurate to within plus or minus 1 percent.

4.4.3.11 Saw-tooth generator, capable of delivering 30 volts peak into a 10,000 ohm load with sweep rate adjustable from 6 Hz to 30 Hz.

4.4.4 Test procedure

4.4.4.1 Dielectric strength. Apply the ac power source (4.4.3.1) across the contact terminals, shorted together, and the relay case, for a period of one minute plus or minus five seconds. Apply this power source across contact terminals and a coil terminal for a like period. The relay shall meet the requirements specified in 3.5.1.

4.4.4.2 Insulation resistance. With the relay deenergized, measure with the megohmmeter (4.4.3.2) or bridge (4.4.3.3) the insulation resistance between the following points:

(a) Relay contact terminals, shorted together, and the relay case.

(b) Contact terminals, shorted together, and a coil terminal.

The relay shall meet the requirements of 3.5.2.

4.4.4.3 Contact resistance. With the relay deenergized, measure with the Kelvin bridge or its equivalent, the resistance across the normally closed contacts. With the rated current of two amperes flowing through the normally closed contacts, measure the resistance across each pair of contacts. With the relay coil energized to close the normally open contacts and with rated current flowing through the contacts, measure the resistance across each pair of contacts. The contact resistance shall be as specified in 3.5.3. Upon completion of the operating life test (4.5.5), the contact resistance, measured as above, shall be no more than double the value specified in 3.5.3.

4.4.4.4 Contact bounce. With the oscilloscope (4.4.3.7), test the relay to determine whether or not the relay exhibits any evidence of chatter or contact bounce. The relay shall meet the requirements of 3.5.4.

4.4.4.5 Coil resistance. Measure the coil resistance of the relay, using any suitable means. The coil resistance shall lie within the limits specified in 3.5.5 when corrected to the temperature indicated.

4.4.4.6 Coil current. With the relay deenergized, check continuity between each pair of normally closed contacts, using the dc voltage source

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(4.4.3.4) and voltmeter (4.4.3.6). With the coil energized by a coil current of 2.8 ma, check continuity between each pair of normally open contacts. Continuity shall exist as indicated on applicable drawings (see 3.4.5.3).

4.4.4.6.1 Operate. With rated current flowing through the set of normally closed contact terminals, apply the voltage source of 4.4.3.4 to a load consisting of the relay coil in series with the milliammeter (4.4.3.9) and the variable resistance (4.4.3.5) set at maximum resistance. Gradually decrease this resistance and determine the point at which the contacts transfer. Record the current value. The current value at which the contacts transfer shall be not greater than that specified in 3.5.6.1.

4.4.4.6.2 Release. With the circuit arrangement of 4.4.4.6.1 and the relay coil energized with a current of 2.8 ma, gradually increase the resistance value until contacts transfer. The current value at which the contacts transfer shall be not less than that specified in 3.5.6.2.

4.4.4.7 Sensitivity. From the values determined in 4.4.4.5 and 4.4.4.6, compute the operating and releasing sensitivity of the relay. Sensitivity shall be as specified in 3.5.7.

4.4.4.8 Operating and releasing time. With the oscilloscope (4.4.3.7) or the saw tooth generator (4.4.3.8), test the relay for chatter and contact bounce and determine the operating and releasing time of the relay when operating at a rate of six transfers per second. The relay shall meet the requirements specified in 3.5.4 and 3.5.8.

4.4.4.9 Transfer time on operate. Adjust the output voltage and sweep rate of the saw tooth generator (4.4.3.11) to obtain a voltage output rate of a maximum 15 volts per second and a minimum peak voltage of 20 volts, using the oscilloscope (4.4.3.7). Apply this voltage to the relay coil and determine the transfer time on operate with the oscilloscope (4.4.3.7). This time is the total elapsed time from the instant the normally closed contacts start to open until the normally open contacts are closed and all contact bounce has ceased. This time shall be less than 6 milliseconds.

4.4.4.10 Visual and dimensional examination. Each relay shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.4.4.11 Packaging, packing, and marking. Packaging, packing, and marking shall comply with the requirements of Section 5.

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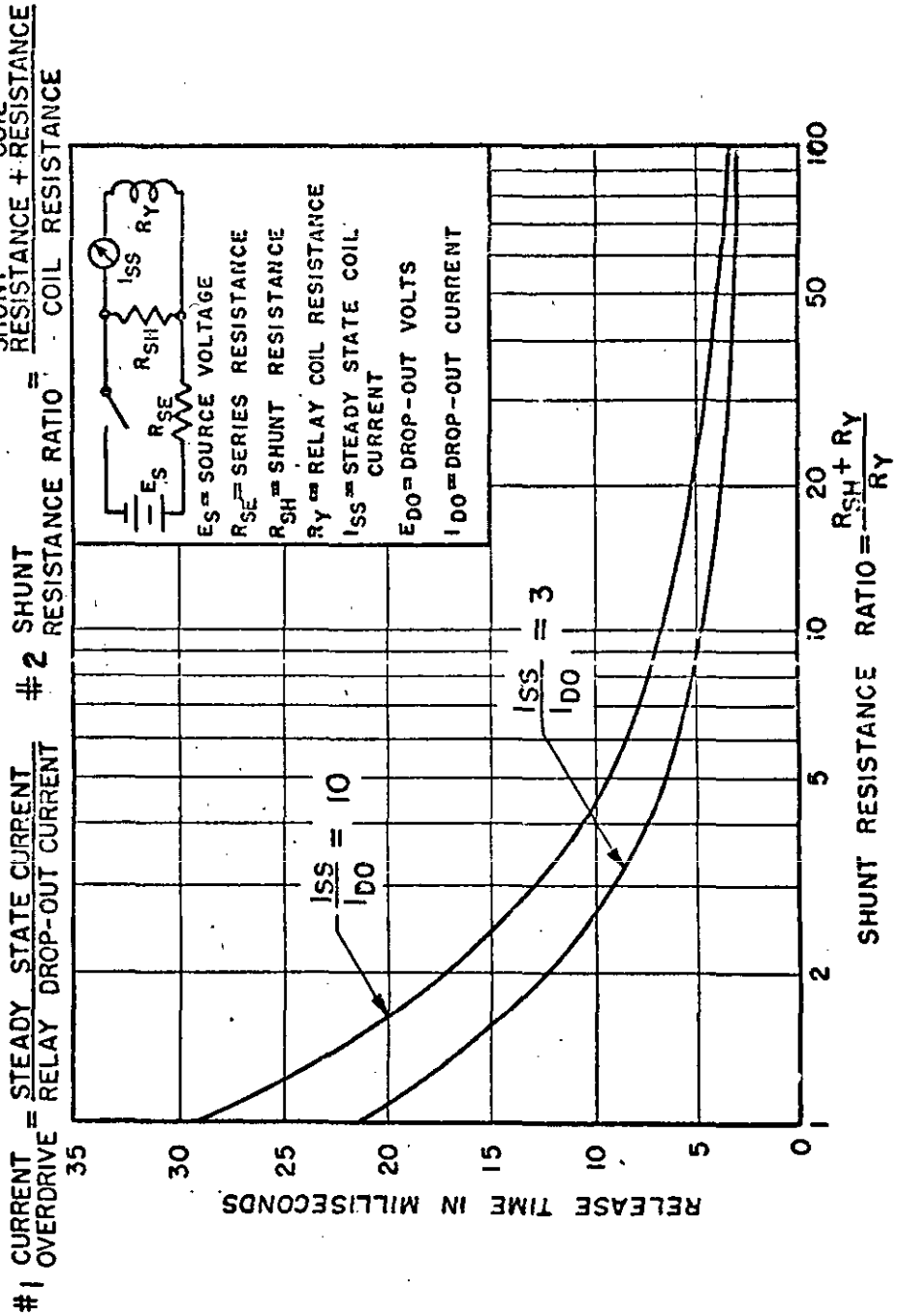


FIGURE-2

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4.5 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.4, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.5.1 Vibration

4.5.1.1 Mounting. The relay, less pack or packaging, shall be mounted securely to a vibration device by means of its mounting holes and tested as specified in 4.5.1.2 and 4.5.1.3.

4.5.1.2 Vibration, nonoperating. The relay shall be subjected to the "Vibration, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.6.1.1.

4.5.1.3 Vibration, operating. The relay shall be subjected to the "Vibration, operating" test of MIL-E-82590 and during and thereafter shall meet the requirements of 3.6.1.2.

4.5.1.4 Vibration, transportation. The relay shall be subjected to the "Vibration, transportation" test of MIL-E-82590 and thereafter shall meet the requirements of 3.6.1.3.

4.5.2 Shock, transportation. The relay shall be subjected to the "Shock, transportation" test of MIL-E-82590 and thereafter shall meet the requirements of 3.6.2.

4.5.3 Temperature and humidity

4.5.3.1 High temperature and humidity, nonoperating. The relay shall be subjected to "High temperature and humidity, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.6.3.1.

4.5.3.2 High temperature, operating. The relay shall be subjected to the "High temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.6.3.2.

4.5.3.3 Low temperature, nonoperating. The relay shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.6.3.3.

4.5.3.4 Low temperature, operating. The relay shall be subjected to the "Low temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.6.3.4

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4.5.4 Pressure. The relay shall be subjected to pressure testing as specified in 3.6.4 and as defined in "Pressure Test Procedure, Submarine Launched" in Section 4 of MIL-T-18404. At the conclusion of the test, the relay shall meet the performance requirements and product characteristics of this specification.

4.5.5 Operating life. Operate the relay, under the environmental conditions described in 4.4.1, for a total of 100,000 contact transfers while the contacts are carrying the rated current of 2 amperes. At the conclusion of the test the relay shall meet the performance requirements and product characteristics of this specification.

4.6 Waiver of tests. Tests specified in 4.5.1.4 and 4.5.2 above may be waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.7 Functional tests. At the conclusion of the tests specified in 4.5.1.4 and 4.5.2 above, functional and other tests, as applicable, shall be conducted on the relay to determine freedom from operational malfunction and compliance with all performance requirements and product characteristics of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging. The relay shall be packaged in accordance with Method III of MIL-P-116.

5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Unit packaging. The relay shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.2.1 Level A

5.2.1.1 Consolidation container. The relays, packaged as described in 5.1.1.1, shall be packed in quantities as specified in the contract or order, in a wooden box in accordance with PPP-B-621, Class 2.

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5.2.1.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Pattern No. 95.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The relay, packaged in accordance with 5.1.3.1, shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Levels A or C cushioning. The unit package shall incorporate sufficient cushioning material, bracing or other adequate shock absorbing devices to ensure that the relay will meet all the requirements of Section 3 after being tested in accordance with 4.5.1.4 (Vibration) and 4.5.2 (Shock). At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent.

5.4 Marking

5.4.1 Special markings. None, unless otherwise specified.

5.4.2 Normal markings. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended use. The sensitive relay is intended for use in Torpedo Mk 37 Mods 0, 2, and 3.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification.

(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing, including palletizing (see 5.2).

(e) Assigned activity for preproduction and periodic production inspections.

(f) Data required other than that specified in 3.2.1.

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(g) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

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